Butyrylcholinesterase activity in patients with postoperative delirium after cardiothoracic surgery or percutaneous valve replacement- an observational interdisciplinary cohort study.

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Postoperative delirium is a frequent and severe complication after cardiac surgery. Activity of butyrylcholinesterase (BChE) has been discussed controversially regarding a possible role in its development. This study aimed to investigate the relevance of BChE activity as a biomarker for postoperative delirium after cardiac surgery or percutaneous valve replacement. A total of 237 patients who received elective cardiothoracic surgery or percutaneous valve replacement at a tertiary care centre were admitted preoperatively. These patients were tested with the Montreal Cognitive Assessment investigating cognitive deficits, and assessed for postoperative delirium twice daily for three days via the 3D-CAM or the CAM-ICU, depending on their level of consciousness. BChE activity was measured at three defined time points before and after surgery. Postoperative delirium occurred in 39.7% of patients ($n = 94$). Univariate analysis showed an association of pre- and postoperative BChE activity with its occurrence ($p = 0.037$, $p = 0.001$). There was no association of postoperative delirium and the decline in BChE activity (pre- to postoperative, $p = 0.327$). Multivariable analysis including either preoperative or postoperative BChE activity as well as age, MoCA, type 2 diabetes mellitus, coronary heart disease, type of surgery and intraoperative administration of red-cell concentrates was performed. Neither preoperative nor postoperative BChE activity was independently associated with the occurrence of postoperative delirium ($p = 0.086$, $p = 0.484$). Preoperative BChE activity was lower in older patients ($B = -12.38$ (95% CI: -21.94 to -2.83), $p = 0.011$), and in those with a history of stroke ($B = -516.173$ (95% CI: -893.927 to -138.420), $p = 0.008$) or alcohol abuse ($B = -451.47$ (95% CI: -868.38 to -34.55), $p = 0.034$). Lower postoperative BChE activity was independently associated with longer procedures ($B = -461.90$ (95% CI: -166.34 to -757.46), $p = 0.002$), use of cardiopulmonary bypass ($B = -262.04$ (95% CI: -485.68 to -38.39), $p = 0.022$), the number of administered red cell-concentrates ($B = -40.99$ (95% CI: -67.86 to -14.12), $p = 0.003$) and older age ($B = -9.35$ (95% CI: -16.04 to -2.66), $p = 0.006$). BChE activity is not independently associated with the occurrence of postoperative delirium. Preoperative BChE values are related to patients' morbidity and vulnerability, while postoperative activities reflect the severity, length and complications of surgery.

Delirium and cardiac surgery: progress - and more questions.

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Post-operative delirium is a common and dangerous complication of cardiac surgery. Many risk factors for delirium have been identified, but its pathogenesis remains largely elusive. A study by Kazmierski and colleagues investigates a more recently considered risk factor for delirium: perturbations in the hypothalamic pituitary axis and depression. This and further work may help define novel prevention and treatment strategies for delirium.

[Risk Factors for Postoperative Delirium after Cardiac Surgery].

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Background Delirium is a common psychiatric disorder after cardiac surgery and predisposes patients to increased mortality and morbidity. Its prevention requires knowledge of the risk factors involved. Objective What are preoperative risk factors for postoperative delirium after cardiac surgery? Methods Prospective longitudinal study of 241 elective cardiac surgical patients with preoperative assessment of potential risk factors and delirium assessment twice daily over five postoperative days. Results 13% of the patients experienced delirium. Reduced cognitive performance (OR: 3.80; 95% CI: 1.66-8.66), higher comorbidity (OR: 1.36; 95% CI: 1.07-1.7) and higher age (OR: 1.08; 95% CI: 1.02-1.13) increased the risk of delirium. Conclusion Delirium after cardiac surgery is common. It occurs in particular in patients with low cognitive performance, higher comorbidity and higher age.

Independent Predictors of the Duration and Overall Burden of Postoperative Delirium After Cardiac Surgery in Adults: An Observational Cohort Study.

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Postoperative delirium (POD) is a common complication after cardiac surgery and is associated with increased patient morbidity and mortality. The objective of this study was to identify risk factors for long duration and overall burden of POD after cardiac surgery. One-year, single-center, retrospective, observational cohort study. University hospital. Adult patients undergoing cardiac surgery with cardiopulmonary bypass in 2013. None. Patients were screened for POD using the Intensive Care Delirium Screening Checklist. The primary outcome measure was the incidence of POD. Secondary outcome measures were the duration of POD and the area under the curve determined using the Intensive Care Delirium Screening Checklist score over time. Independent predictors of POD were estimated in multivariable logistic regression models. Hospital length of stay, medications, and outcome data also were analyzed. Among the 656 patients included in the cohort, 618 were analyzed. The overall incidence of POD was 39%. Older patient age (odds ratio [95% confidence interval]) 1.06 [1.04-1.09] for an increase of 1 year, $p < 0.001$); low preoperative serum albumin (1.08 [1.03-1.13] for a decrease of 1 g/L, $p < 0.001$); a history of atrial fibrillation (2.30 [1.30-4.09], $p = 0.004$); perioperative stroke (6.27 [1.54-43.64], $p = 0.008$); ascending aortic replacement surgery (2.99 [1.50-6.05], $p = 0.002$); longer duration of procedure (1.37 [1.16-1.63] for an increase of 1 hour, $p < 0.001$); and increased postoperative C-reactive protein concentration (2.16 [1.49-3.16] for a 2-fold increase, $p < 0.001$) were associated with higher odds of POD. Among patients affected by POD, older age, perioperative stroke, longer procedure time, and increased postoperative C-reactive protein were consistently predictive of longer duration of POD and greater area under the curve. Known risk factors for the development of POD after cardiac surgery also are predictive of prolonged duration and high overall burden of POD.

Predictors and clinical outcomes of postoperative delirium after administration of dexamethasone in patients undergoing coronary artery bypass surgery.

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Postoperative delirium (POD) is one of the important complications of cardiac surgery and it is assumed to provoke inflammatory responses. Theoretically, anti-inflammatory effects of dexamethasone can have an influence on the incidence and outcomes of POD. The aim of our study was to assess POD predictors and outcomes of dexamethasone administration after cardiac surgery. Patients' mental status was examined by mini-mental status examination and psychiatric interviewing to diagnose delirium. Subsequently, authors analyzed the patient variables for identification of predictors and outcomes of POD. Between 196 patients who met the inclusion criteria, 34 (17.34%) patients were delirious. History of chronic renal failure, obstructive pulmonary disease, smoking, and addiction strongly predicted development of POD. Other predictors were intra-aortic balloon pump insertion, transfusion of packed cells, and atrial fibrillation rhythm. In our study, the administration of dexamethasone significantly reduced the risk for POD. Furthermore, delirium was associated with longer intensive care unit (ICU) stay. Our study reports the predictors of POD, which patients commonly facing them in cardiac surgery ICU. Appropriate management and prevention of these predictors, especially modifiable ones, can decrease the incident of POD and improves cognitive outcomes of cardiac surgeries.

Unveiling the nexus of postoperative fever and delirium in cardiac surgery: identifying predictors for enhanced patient care.

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Postoperative delirium (POD) is a significant complication observed in cardiac surgery patients, characterized by acute cognitive decline, fluctuating mental status, consciousness impairment, and confusion. Despite its impact, POD often goes undiagnosed. Postoperative fever, a common occurrence after cardiac surgery, has not been comprehensively studied in relation to delirium. This study aims to identify perioperative period factors associated with POD in patients undergoing cardiopulmonary bypass, with the potential for implementing preventive interventions. In a prospective observational study conducted between February 2023 and April 2023 at the Department of Cardio-Thoracic Surgery, Nanjing Drum Tower Hospital, Affiliated Hospital of Nanjing University Medical School, a total of 232 patients who underwent cardiac surgery were enrolled. POD assessment utilized the Confusion Assessment Method for the ICU (CAM-ICU), while high fever was defined as a bladder temperature exceeding 39°C. Statistical analysis included univariate and multivariate analyses, logistic regression, nomogram development, and internal validation. The overall incidence of postoperative delirium was found to be 12.1%. Multivariate analysis revealed that postoperative lactate levels [odds ratio (OR) = 1.787], maximum temperature (OR = 11.290), and cardiopulmonary bypass time (OR = 1.015) were independent predictors of POD. A predictive nomogram for POD was developed based on these three factors, demonstrating good discrimination and calibration. The prediction model exhibited a C-statistic value of 0.852 (95% CI, 0.763-0.941), demonstrating excellent discriminatory power. Sensitivity and specificity, based on the area under the receiver operating characteristic (AUROC) curve, were 91.2% and 67.9%, respectively. This study underscores the high prevalence of POD in cardiac surgery patients and identifies postoperative lactate levels, cardiopulmonary bypass duration, and postoperative fever as independent predictors of delirium. The association between postoperative fever and POD warrants further investigation. These findings have implications for implementing preventive strategies in high-risk patients, aiming to mitigate postoperative complications and improve patient outcomes.

Electroencephalography Monitoring for Preventing Postoperative Delirium and Postoperative Cognitive Decline in Patients Undergoing Cardiothoracic Surgery: A Meta-Analysis.

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Patients undergoing cardiothoracic surgery frequently encounter perioperative neurocognitive disorders (PND), which can include postoperative delirium (POD) and postoperative cognitive decline (POCD). Currently, there is not enough evidence to support the use of electroencephalograms (EEGs) in preventing POD and POCD among cardiothoracic surgery patients. This meta-analysis examined the importance of EEG monitoring in POD and POCD. Cochrane Library, PubMed, and EMBASE databases were searched to obtain the relevant literature. This analysis identified trials based on the inclusion and exclusion criteria. The Cochrane tool was used to evaluate the methodological quality of the included studies. Review Manager software (version 5.3) was applied to analyze the data. Four randomized controlled trials (RCTs) were included in this meta-analysis, with 1096 participants. Our results found no correlation between EEG monitoring and lower POD risk (relative risk (RR): 0.81; 95% CI: 0.55-1.18; $p = 0.270$). There was also no statistically significant difference between the EEG group and the control group in the red cell transfusions (RR: 0.86; 95% CI: 0.51-1.46; $p = 0.590$), intensive care unit (ICU) stay (mean deviation (MD): -0.46; 95% CI: -1.53-0.62; $p = 0.410$), hospital stay (MD: -0.27; 95% CI: -2.00-1.47; $p = 0.760$), and mortality (RR: 0.33; 95% CI: 0.03-3.59; $p = 0.360$). Only one trial reported an incidence of POCD, meaning we did not conduct data analysis on POCD risk. This meta-analysis did not find evidence supporting EEG monitoring as a potential method to reduce POD incidence in cardiothoracic surgery patients. In the future, more high-quality RCTs with larger sample sizes are needed to validate the relationship between EEG monitoring and POD/POCD further.

Risk Factors for Acute Postoperative Delirium in Cardiac Surgery Patients >65 Years Old.

PMID: 36143313 | Indexed in PubMed

Background: Acute postoperative delirium is the most common neuropsychiatric disorder in cardiac surgery patients in the intensive care unit (ICU). The purpose of this study was to evaluate the possible risk factors of postoperative delirium (POD) for cardiac surgery patients in the ICU. Materials and Methods: The study population was composed of 86 cardiac surgery patients managed postoperatively in the cardiac surgery ICU. Presence of POD in patients was evaluated by the CAM-ICU scale. Results: According to the CAM-ICU scale, 22 (25.6%) patients presented POD; history of smoking, alcohol use, COPD, and preoperative permanent atrial fibrillation were associated with POD (for all, p 0.05), while cardiac arrhythmia in the ICU, hypoxemia in the ICU after extubation ($pO_2 < 60$ mmHg), and heart rate after extubation were predisposing factors for POD (for all, $p < 0.05$). Multivariable logistic regression analysis (adjusted to risk factors) showed that hypoxemia after extubation (OR = 20.6; 95%CI: 2.82−150), heart rate after extubation (OR = 0.95; 95% CI: 0.92−0.98), and alcohol use (OR = 74.3; 95%CI: 6.41−861) were predictive factors for acute postoperative delirium (for all, $p < 0.05$). Conclusion: Alcohol use and respiratory dysfunction before and after heart operation were associated with acute postoperative delirium in cardiac surgery ICU patients.

Intra-operative electroencephalogram frontal alpha-band spectral analysis and postoperative delirium in cardiac surgery: A prospective cohort study.

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Postoperative delirium (POD) remains a frequent complication after cardiac surgery, with pre-operative cognitive status being one of the main predisposing factors. However, performing complete pre-operative neuropsychological testing is challenging. The magnitude of frontal electroencephalographic (EEG) α oscillations during general anaesthesia has been related to pre-operative cognition and could constitute a functional marker for brain vulnerability. We hypothesised that features of intra-operative α-band activity could predict the occurrence of POD. Single-centre prospective observational study. University hospital, from 15 May 2019 to 15 December 2021. Adult patients undergoing elective cardiac surgery. Pre-operative cognitive status was assessed by neuropsychological tests and scored as a global z score. A 5-min EEG recording was obtained 30 min after induction of anaesthesia. Anaesthesia was maintained with sevoflurane. Power and peak frequency in the α-band were extracted from the frequency spectra. POD was assessed using the Confusion Assessment Method for Intensive Care Unit, the Confusion Assessment Method and a chart review. Sixty-five (29.5%) of 220 patients developed POD. Delirious patients were significantly older with median [IQR] ages of 74 [64 to 79] years vs. 67 [59 to 74] years; $P < 0.001$) and had lower pre-operative cognitive z scores (-0.52 ± 1.14 vs. 0.21 ± 0.84; $P < 0.001$). Mean α power (-14.03 ± 4.61 dB vs. -11.59 ± 3.37 dB; $P < 0.001$) and maximum α power (-11.36 ± 5.28 dB vs. -8.85 ± 3.90 dB; $P < 0.001$) were significantly lower in delirious patients. Intra-operative mean α power was significantly associated with the probability of developing POD (adjusted odds ratio, 0.88; 95% confidence interval (CI), 0.81 to 0.96; $P = 0.007$), independently of age and only whenever cognitive status was not considered. A lower intra-operative frontal α-band power is associated with a higher incidence of POD after cardiac surgery. Intra-operative measures of α power could constitute a means of identifying patients at risk of this complication. NCT03706989.

Diagnosis, prevention, and management of delirium in the intensive cardiac care unit.

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Delirium is a frequent complication in patients admitted to intensive cardiac care units (ICCU) with potentially severe consequences including increased risks of mortality, cognitive impairment and dependence at discharge, and longer times on mechanical ventilation and hospital stay. Delirium has been widely documented and studied in general intensive care units and in patients after cardiac surgery, but it has barely been studied in acute nonsurgical cardiac patients. Moreover, delirium (especially in its hypoactive form) is commonly misdiagnosed. We propose a protocol for delirium prevention and management in ICCUs. A daily comprehensive assessment to improve detection should be done using validated scales (ie, confusion assessment method). Preventive measures are particularly relevance and constitute the basis of treatment as well, acting on reversible risk factors, including environmental interventions, such as quiet time, sleep promotion, family support, communication, and adequate treatment of pain and dyspnea. Pharmacological prophylaxis is not indicated with the exception of patients at risk of withdrawal syndrome but should only be used in patients with confirmed delirium. Dexmedetomidine is the drug of choice in patients with severe agitation, and those weaning from invasive mechanical ventilation. As the complexity of ICCUs increases, clinical scenarios posing challenges for the management of delirium become more frequent. Efforts should be done to improve the identification of patients at risk during admission in order to establish preventive interventions to avoid this complication. Patient-centered protocols will increase the awareness of the healthcare professionals for better prevention and earlier diagnosis and will positively impact on prognosis.